Quantifying Uncertainty with MR-PhICC:

MRI Reconstruction with Physics-Informed Conformal Calibration

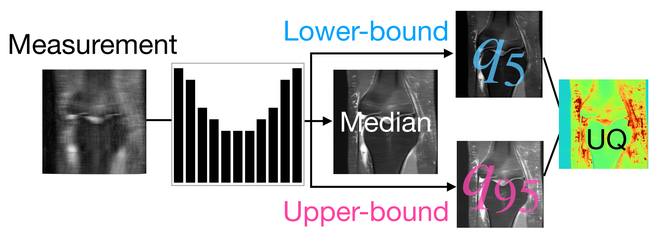
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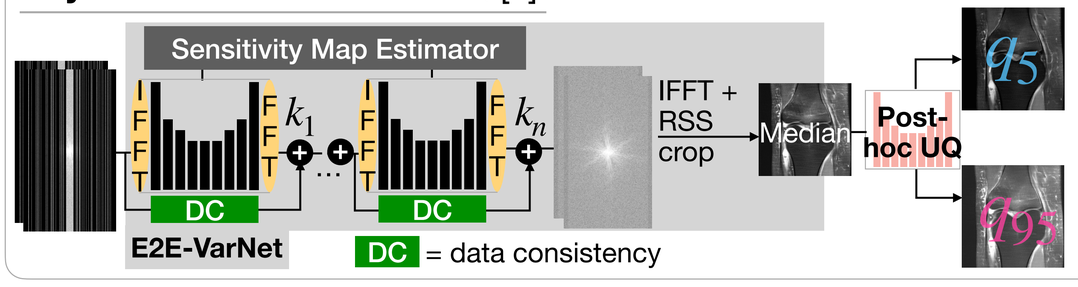
Motivation: Machine-learning priors in MRI reconstruction methods risk introducing **hallucinations**; uncertainty quantification methods (UQ) provide statistically bounds on reconstruction uncertainty.

Deep learning only: im2im [1]

Methods like CNNs or transformers are not physics-based.



Physics-based: Pixelwise UQ [2]



Physics-based methods enforce measurement fidelity (data consistency). However, current UQ methods are post-hoc and not integrated with the reconstruction.

Goal: An integrated approach to uncertainty quantification for physics-based architectures

Method: Integrated UQ with qVarNet and conformal calibration

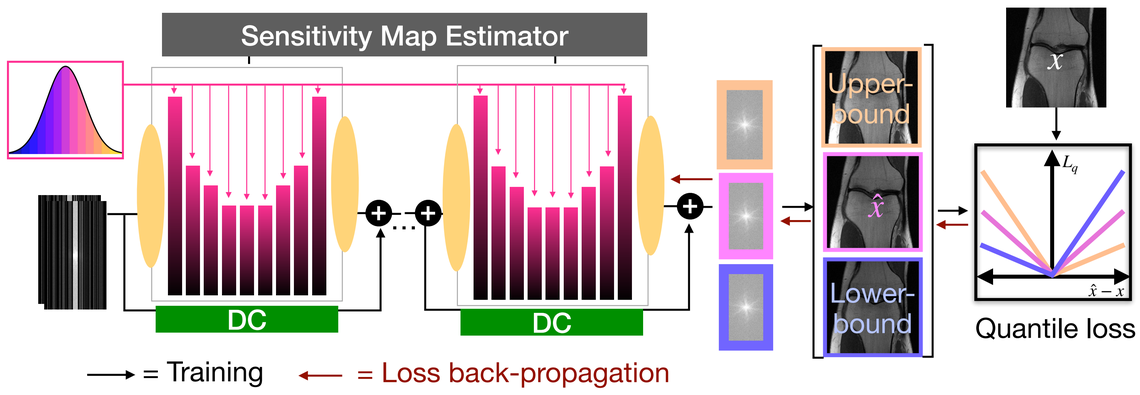
Quantile loss

- Prior work [1,2] reconstructs only fixed quantiles using the quantile loss function:

$$L_q(x,\hat{x}) = \begin{cases} q\,|x-\hat{x}| & \text{if } x-\hat{x} \geq 0 \\ (1-q)\,|x-\hat{x}| & \text{otherwise} \end{cases}$$

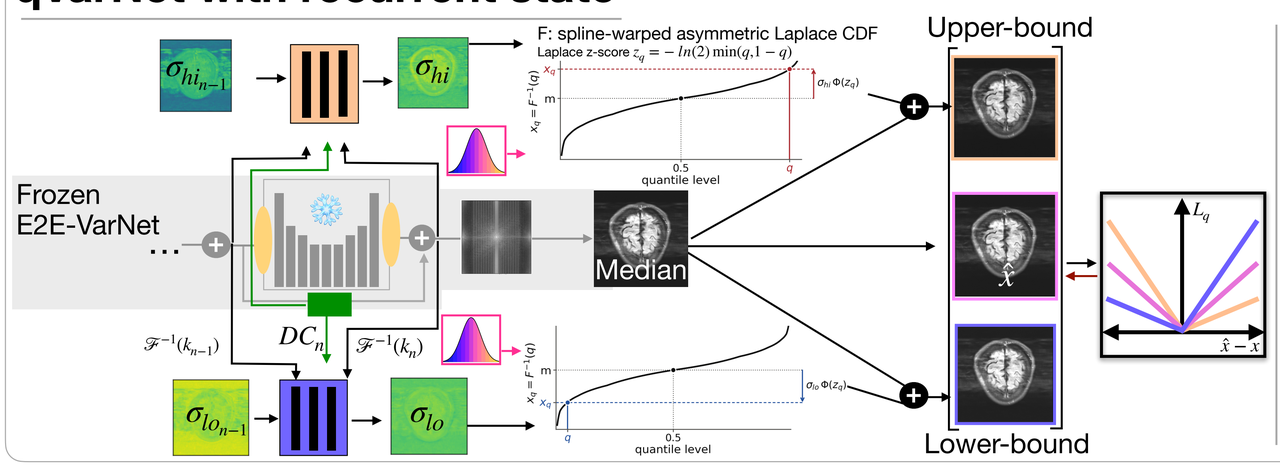
- Instead, we use quantile loss to condition the reconstruction on any quantile, similarly to [3]

qVarNet with quantile conditioning



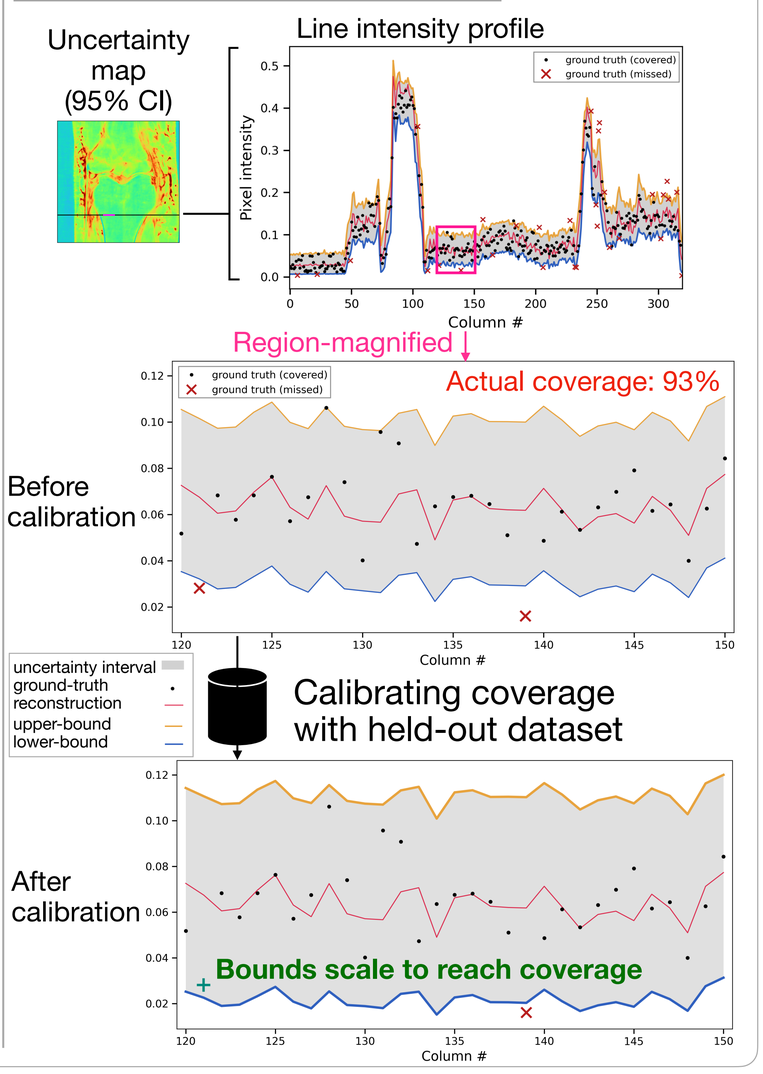
- We add FiLM layers [4] within regularizer blocks to modulate neural-network layers conditional on quantile embeddings
- SSIM loss: median, quantile loss: quantile reconstructions

qVarNet with recurrent state

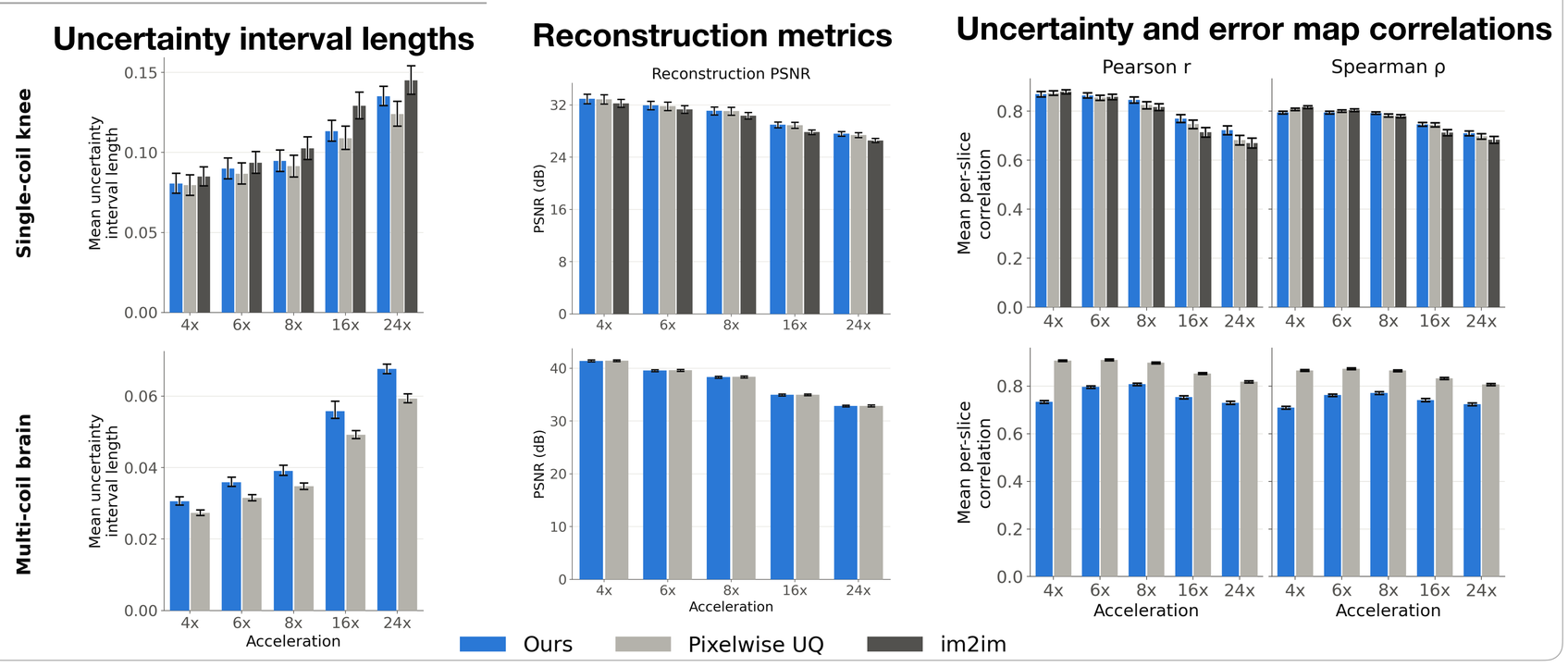


- Quantile conditioning training is expensive with multi-coil data
- Alternatively, we train UQ modulators from frozen reconstruction
- Quantiles computed as sum of median and weighted modulators [5]

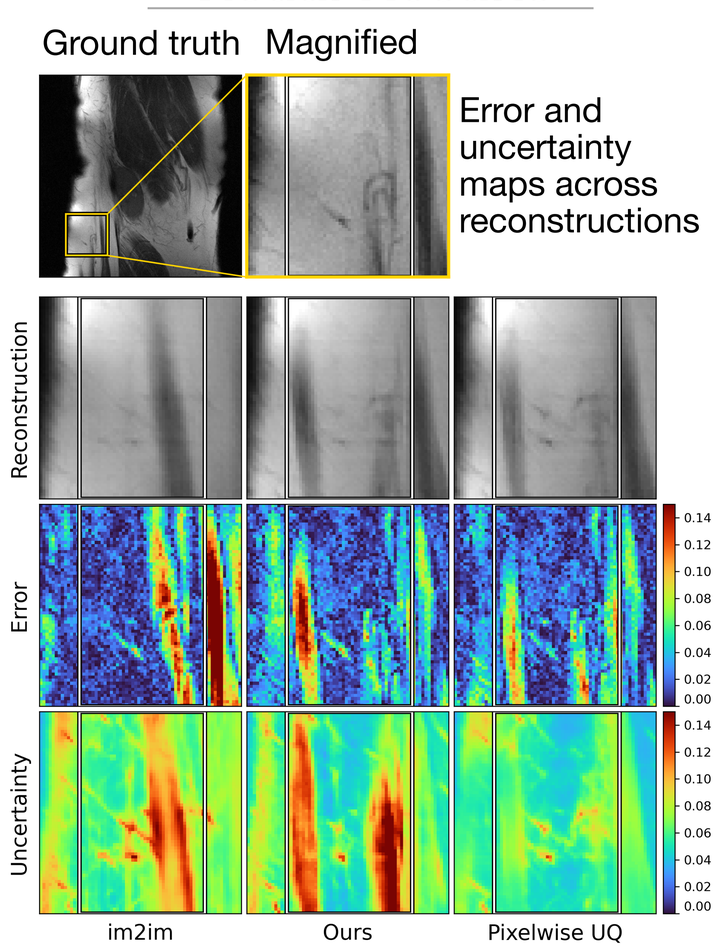
Conformal calibration



Quantitative Results



Visual Results



Further exploration

- qVarNet focused on quantile-conditioning for the regularizer, but how does that optimization interact with the fixed data-consistency constraints?
- Can integrated UQ provide tighter bounds versus a post-hoc approach when training data is limited?

Conclusion:

✅ Physics-based reconstruction methods out-perform deep-learning-only ones.
❓ Surprisingly, integrated UQ performs comparably to post-hoc approaches.

References:
[1] Angelopoulos, A.N., Kohli, A.P., Bates, S., Jordan, M., Malik, J., Alshaabi, T., Upadhyayula, S. & Romano, Y. (2022). Image-to-Image Regression with Distribution-Free Uncertainty Quantification and Applications in Imaging. *Proceedings of the 39th Int'l Conf. on Machine Learning*. [2] I. I. Giannakopoulos, L. B. Gautham Muthukumar, Y. W. Lui, and R. Lattanzi, "Pixelwise uncertainty quantification of accelerated MRI reconstruction," Jan. 2026. [3] Ye, C. T., Li, S., King, T., & Monakhova, K. (2025). QUTCC: Quantile Uncertainty Training and Conformal Calibration for Imaging Inverse Problems. [4] Perez, E., Strub, F., de Vries, H., Dumoulin, V., & Courville, A. C. (2018). FiLM: Visual Reasoning with a General Conditioning Layer. *AAAI*. [5] Yu, K., & Moyeed, R. A. (2001). Bayesian quantile regression. *Statistics & Probability Letters*. [6] Bates, S., Angelopoulos, A., Lei, L., Malik, J., & Jordan, M. (2021). Distribution-free, Risk-controlling Prediction Sets. J. ACM.

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